

DevOps Lab Programs

Program10: Demonstrate minikube dashboard using CLI and kubernetes

Project structure:

Program10/

→deployment.yaml

→service.yaml

Steps for execution:

Step 1: start minikube and check the status of minikube

Step 2: Create deployment.yaml file with the following code



```
GNU nano 8.1
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

Step 3: Create services.yaml file with the following code

```
GNU nano 8.1
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort
```

Step 4: Apply the files

Kubectl apply -f deployment.yaml

Kubectl apply -f services.yaml

Step 5: Get pods, deployment, services

Kubectl get pods

Kubectl get services

Kubectl get deployment

```
Dec 31 15:09
triv24mc083-rakshitthadas@triv24mc083rakshitthadas:~/Docker/lab10
triv24mc083-rakshitthadas@triv24mc083rakshitthadas:~/Docker/lab10
triv24mc083-rakshitthadas@triv24mc083rakshitthadas:~/Docker/lab10
triv24mc083-rakshitthadas@triv24mc083rakshitthadas:~/Docker/lab10$ nano deployment.yaml
triv24mc083-rakshitthadas@triv24mc083rakshitthadas:~/Docker/lab10$ nano service.yaml
triv24mc083-rakshitthadas@triv24mc083rakshitthadas:~/Docker/lab10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment unchanged
triv24mc083-rakshitthadas@triv24mc083rakshitthadas:~/Docker/lab10$ kubectl apply -f service.yaml
service/nginx-service unchanged
triv24mc083-rakshitthadas@triv24mc083rakshitthadas:~/Docker/lab10$ kubectl get pods
NAME                                READY    STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-slgf7   1/1      Running   1 (5m25s ago)    6m1s
nginx-deployment-6f9664446b-d5nqe   1/1      Running   1 (5m25s ago)    6m1s
triv24mc083-rakshitthadas@triv24mc083rakshitthadas:~/Docker/lab10$ kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
minikube                            Ready     control-plane   23s    v1.34.0
triv24mc083-rakshitthadas@triv24mc083rakshitthadas:~/Docker/lab10$ kubectl get svc
NAME                                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
kubernetes                          ClusterIP   10.96.0.1      <none>          443/TCP    23d
kubia                                NodePort    10.106.174.127 <none>          8080:32481/TCP    23d
nginx-service                       NodePort    10.97.155.196  <none>          80:38866/TCP    19h
triv24mc083-rakshitthadas@triv24mc083rakshitthadas:~/Docker/lab10$ kubectl logs nginx-deployment-6f9664446b-slgf7
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: Info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: Info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-time-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/12/31 09:27:45 [notice] 1#1: using the "epoll" event method
2025/12/31 09:27:45 [notice] 1#1: nginx/1.29.4
2025/12/31 09:27:45 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2025/12/31 09:27:45 [notice] 1#1: OS: Linux 6.14.0-33-generic
2025/12/31 09:27:45 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/12/31 09:27:45 [notice] 1#1: start worker processes
2025/12/31 09:27:45 [notice] 1#1: start worker process 29
2025/12/31 09:27:45 [notice] 1#1: start worker process 30
2025/12/31 09:27:45 [notice] 1#1: start worker process 31
2025/12/31 09:27:45 [notice] 1#1: start worker process 32
2025/12/31 09:27:45 [notice] 1#1: start worker process 33
2025/12/31 09:27:45 [notice] 1#1: start worker process 34
2025/12/31 09:27:45 [notice] 1#1: start worker process 35
2025/12/31 09:27:45 [notice] 1#1: start worker process 36
triv24mc083-rakshitthadas@triv24mc083rakshitthadas:~/Docker/lab10$ kubectl describe pod nginx-deployment-6f9664446b-slgf7
```

Step 6: Describe the pod

```
1rv24mc083-rakshithadas@1rv24mc083rakshithadas: ~/Docker/lab10
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/Lab10$ nano deployment.yaml
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/Lab10$ nano service.yaml
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/Lab10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment unchanged
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/Lab10$ kubectl apply -f service.yaml
service/nginx-service unchanged
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/Lab10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-5lqf7   1/1     Running   1 (5n25s ago)    6m1s
nginx-deployment-6f9664446b-d5nqw   1/1     Running   1 (5n25s ago)    6m1s
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/Lab10$ kubectl get nodes
NAME                STATUS   ROLES    AGE   VERSION
minikube            Ready    control-plane  23d   v1.34.0
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/Lab10$ kubectl get svc
NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1     <none>         443/TCP           23d
kubia               NodePort    10.106.174.127 <none>         8080:32481/TCP   23d
nginx-service       NodePort    10.97.155.196 <none>         80:30866/TCP     19h
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/Lab10$ kubectl logs nginx-deployment-6f9664446b-5lqf7
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/12/31 09:27:45 [notice] 1#1: using the "epoll" event method
2025/12/31 09:27:45 [notice] 1#1: nginx/1.29.4
2025/12/31 09:27:45 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2025/12/31 09:27:45 [notice] 1#1: OS: Linux 6.14.0-33-generic
2025/12/31 09:27:45 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
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2025/12/31 09:27:45 [notice] 1#1: start worker process 35
2025/12/31 09:27:45 [notice] 1#1: start worker process 36
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/Lab10$ kubectl describe pod nginx-deployment-6f9664446b-5lqf7
Name:          nginx-deployment-6f9664446b-5lqf7
Namespace:     default
Priority:       0
Service Account: default
Node:          minikube/192.168.49.2
Start Time:    Wed, 31 Dec 2025 14:55:40 +0530
Labels:        app=nginx
               pod-template-hash=6f9664446b
Annotations:   <none>
Status:        Running
IP:            10.244.0.75
IPs:           IP: 10.244.0.75
               Controlled By: ReplicaSet/nginx-deployment-6f9664446b
Containers:
  nginx:
    Container ID:  docker://c96e9443bd0a5e6b9505fc597bc1dc64986e3d7b17bae6c9ffbab875fc445d2
    Image:          nginx:latest
    Image ID:       docker-pullable://nginx@sha256:c8dc101ba61a499398c245fed8123abc20a3867ebd3664087890a054a781770b
    Port:          80/TCP
    Host Port:     80/TCP
    State:         Running
      Started:     Wed, 31 Dec 2025 14:57:45 +0530
    Last State:    Terminated
      Reason:      Completed
      Exit Code:    0
      Started:     Wed, 31 Dec 2025 14:55:53 +0530
      Finished:    Wed, 31 Dec 2025 14:56:16 +0530
    Ready:         True
    Restart Count:  1
    Environment:   <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-vdc82 (ro)
Conditions:
  Type              Status
  PodReadyToStartContainers  True
  Initialized        True
  Ready              True
  ContainersReady     True
  PodScheduled       True
```

Step 7: Check the minikube services

```
1rv24mc083-rakshithadas@1rv24mc083rakshithadas: ~/Docker/lab10
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/Lab10$ kubectl describe pod nginx-deployment-6f9664446b-5lqf7
Name:          nginx-deployment-6f9664446b-5lqf7
Namespace:     default
Priority:       0
Service Account: default
Node:          minikube/192.168.49.2
Start Time:    Wed, 31 Dec 2025 14:55:40 +0530
Labels:        app=nginx
               pod-template-hash=6f9664446b
Annotations:   <none>
Status:        Running
IP:            10.244.0.75
IPs:           IP: 10.244.0.75
               Controlled By: ReplicaSet/nginx-deployment-6f9664446b
Containers:
  nginx:
    Container ID:  docker://c96e9443bd0a5e6b9505fc597bc1dc64986e3d7b17bae6c9ffbab875fc445d2
    Image:          nginx:latest
    Image ID:       docker-pullable://nginx@sha256:c8dc101ba61a499398c245fed8123abc20a3867ebd3664087890a054a781770b
    Port:          80/TCP
    Host Port:     80/TCP
    State:         Running
      Started:     Wed, 31 Dec 2025 14:57:45 +0530
    Last State:    Terminated
      Reason:      Completed
      Exit Code:    0
      Started:     Wed, 31 Dec 2025 14:55:53 +0530
      Finished:    Wed, 31 Dec 2025 14:56:16 +0530
    Ready:         True
    Restart Count:  1
    Environment:   <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-vdc82 (ro)
Conditions:
  Type              Status
  PodReadyToStartContainers  True
  Initialized        True
  Ready              True
  ContainersReady     True
  PodScheduled       True
```

Step 8: check the status of pods using dashboard

The screenshot displays the Kubernetes Dashboard interface. The top navigation bar shows the 'Workloads' section is active. The left sidebar contains a list of resources: Workloads, Cron Jobs, Daemon Sets, Deployments, Jobs, Pods, Replica Sets, Replication Controllers, Stateful Sets, Service, Ingresses, Ingress Classes, Services, Config and Storage, Config Maps, Persistent Volume Claims, and Secrets. The main content area is divided into two sections. The top section, 'Workload Status', features three large green circles representing the status of Deployments, Pods, and Replica Sets. Below each circle, the text 'Running: 1' is shown for Deployments and Replica Sets, and 'Running: 2' for Pods. The bottom section, 'Deployments', is a table listing the deployment resources.

Name	Images	Labels	Pods	Created
nginx-deployment	nginx:latest	-	2 / 2	19 hours ago